Riemann's Hypothesis as an Eigenvalue Problem

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ABSTRACT

The matrix $A_N = (a_{m,n})_{2 \leqslant m,n \leqslant N}$, where $a_{m,n} = m - 1$ if $m \mid n$ and $a_{m,n} = -1$ if $m \nmid n$, has the determinant $N! \sum_{1 \leqslant n \leqslant N} \mu(n)/n$, μ the Möbius function, and thus is closely connected with the Riemann hypothesis, which is true if and only if $\det A_N = O(N! N^{-1/2+\varepsilon})$ for every $\varepsilon > 0$ (Theorem 1). $\det A_N$ is the product of the eigenvalues λ_n $(1 \leqslant n \leqslant N-1)$ of A_N; therefore it might be of interest to find properties of these eigenvalues. A theorem of Gerschgorin shows immediately $|\lambda_n| \leqslant N - 1/N$ for $1 \leqslant n \leqslant N-1$ (Theorem 2), and a discussion of the characteristic polynomial $\chi_N(x)$ of A_N gives $n \leqslant \lambda_n < n+1$ for all n with at most $N - 2\sqrt{N}$ exceptions. In particular, all natural numbers n such that $N/3 < n \leqslant N/2$ turn out to be eigenvalues of A_N (Theorem 5). The power sums of the eigenvalues have the property

$$\sum_{1 \leqslant n \leqslant N-1} \lambda_n^k \begin{cases} \geqslant 1^k + 2^k + \cdots + (N-1)^k & \text{for all } k \geqslant 0, \\ \leqslant 2^k + 3^k + \cdots + N^k & \text{for } k \leqslant N^{1/2} \text{ and } k \geqslant cN \log^2 N \end{cases}$$

(Theorem 9), so that $g^*_{k,N}$, defined by $\sum_{n=1}^{N-1} \lambda_n^k = \sum_{n=2}^{N-1} n^k + g^*_{k,N} N^k$, is—with regard to $n \leqslant \lambda_n < n+1$—a measure for the uniformity of the distribution of the eigenvalues λ_n in comparison with their indices n. A Tauberian theorem shows that the values $g^*_{k,N}$ almost approach 1 (Corollary of Theorem 8), but $\lim_{k \to \infty} g^*_{k,N} = 0$. If the Riemann hypothesis is true, Theorem 1 might suggest $\lambda_n \sim n + \frac{1}{2}$, but formula (2) in the Corollary of Theorem 11 hints at a slight average shift of the λ_n's to $n+1$. This is exemplified by the largest eigenvalue $\lambda_{\max}$ of A_N (Theorem 11): $\lambda_{\max} = N - [\tau(N)/\zeta_\infty \log N][1 + O(\log_2^2 N/\log N)]$, where $\zeta_\infty = \prod_{k \geqslant 2} \zeta(k) \sim 2.3$, and in $\tau(N) = O(\log_2 N)$ the multiplicative structure of N is recovered.

0. INTRODUCTION

There are many assertions which are equivalent to the Riemann hypothesis (e.g. [4, 14.32], [1, p. 335]). We will add here one more, concerning the

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matrix

$$A_N = (m\delta_{m\mid n} - 1)_{2 \leqslant m,\, n \leqslant N}$$

$$= \begin{pmatrix} 1 & -1 & 1 & -1 & 1 & -1 & 1 & \cdots & \\ & 2 & -1 & -1 & 2 & -1 & -1 & & \\ & & 3 & -1 & -1 & -1 & 3 & & \\ & & & 4 & -1 & -1 & -1 & & \\ & & & & 5 & -1 & -1 & & \\ & & & & & 6 & -1 & & \\ & & -1 & & & & 7 & & \\ & & & & & & & \ddots & \\ & & & & & & & & N-1 \end{pmatrix},$$

where

$$\delta_{m\mid n} = \begin{cases} 1 & \text{for } m \mid n, \\ 0 & \text{for } m \nmid n, \end{cases}$$

and which states:

THEOREM 1. *Riemann's hypothesis is true if and only if*

$$\det A_N = O(N! N^{-1/2+\varepsilon}), \qquad N \to \infty,$$

for every positive ε.

For on the one hand, a simple calculation shows

$$\det A_N = N! \sum_{1 \leqslant n \leqslant N} \frac{\mu(n)}{n}, \qquad \mu \text{ the Möbius function}, \tag{0.1}$$

and on the other hand the Riemann hypothesis is equivalent to the estimate

$$\sum_{1 \leqslant n \leqslant N} \mu(n) = O(N^{1/2+\varepsilon})$$

[4, Theorem 14.25(C)], i.e.—via partial summation—to

$$\sum_{1 \leqslant n \leqslant N} \frac{\mu(n)}{n} = O(N^{-1/2+\varepsilon}).$$

The aim of this paper is to investigate the eigenvalues $\lambda_1, \ldots, \lambda_{N-1}$ of the matrix A_N.

In Section 1 we prove

$$|\lambda_n| \leqslant N - \frac{1}{N} \qquad (1 \leqslant n \leqslant N-1), \tag{0.2}$$

and further, if the eigenvalues are indexed appropriately,

$$n \leqslant \lambda_n < n+1 \tag{0.3}$$

for $1 \leqslant n \leqslant N-1$ with at most $2\sqrt{N}$ exceptions. In particular all natural numbers n in the interval $N/3 < n \leqslant N/2$ turn out to be eigenvalues of A_N. It remains an open question whether eigenvalues of order greater than 1 exist; in any case all eigenspaces are one-dimensional.

Section 2 deals with the power sums

$$\sum_{n=1}^{N-1} \lambda_n^k =: \sum_{n=2}^{N} n^k - g_{k,N} N^k.$$

(0.2) shows directly

$$\lim_{k \to \infty} g_{k,N} = 1,$$

and we prove

$$g_{k,N} \leqslant 1 - \frac{1}{N^k} \qquad \text{for all natural numbers } k,$$

$$g_{k,N} \geqslant 0 \qquad \text{for } k \leqslant N^{1/2} \text{ and } k \geqslant cN \log^2 N.$$

Probably $g_{k,N} \geqslant 0$ is true for all k, and hence

$$1^k + 2^k + \cdots + (N-1)^k \leqslant \sum_{n=1}^{N-1} \lambda_n^k \leqslant 2^k + 3^k + \cdots + N^k,$$

which agrees well with (0.3). In any case the $g_{k,N}$ can become smaller than any positive ε.

Looking at Theorem 1, at the inequalities (0.3), and at

$$\prod_{n=1}^{N-1}\left(n+\tfrac{1}{2}\right) \sim c_1 N! N^{-1/2},$$

one may expect "on average"

$$\lambda_n \sim n+\tfrac{1}{2} \qquad (1 \leqslant n \leqslant N-1).$$

But in Section 3 we give a representation of the characteristic polynomial $\chi_N(x)$ of A_N from which follows

$$\prod_{n=1}^{N-1}(N-\lambda_n) \sim (N-1)! N^{-1}\tau(N), \qquad \tau(N) = O(\log_2 N),$$

contrary to the estimate

$$\prod_{n=1}^{N-1}\left\{N-\left(n+\tfrac{1}{2}\right)\right\} \sim c_2(N-1)! N^{-1/2},$$

which hints at a slight shift of the values λ_n to the right in comparison with $n+\frac{1}{2}$. This can be exemplified by the largest eigenvalue $\lambda_{\max} := \max_{1 \leqslant n \leqslant N-1}|\lambda_n|$, i.e. the spectral radius of A_N:

$$\lambda_{\max} = N - \frac{\tau(N)}{\zeta_\infty \log N}\left\{1+O\left(\frac{\log_2^2 N}{\log N}\right)\right\},$$

where

$$\zeta_\infty = \prod_{k=2}^{\infty}\zeta(k) = 2.294856\ldots,$$

and in

$$\tau(N) = \prod_{p^s \| N}\prod_{1 \leqslant r \leqslant s}(1-p^{-r})^{-1} = O(\log_2 N)$$

the multiplicative structure of N is recovered.

NOTATION.

$\zeta(s)$ is the Riemann zeta function,
$\mu(n)$ is the Möbius function,
p denotes primes,
$p^t \| n$ means: $p^t \mid n$ and $p^{t+1} \nmid n$,
$[y]$ denotes the largest integer $\leqslant y$,
$]a, b] = \{x;\ a < x \leqslant b\}$,
$\mathbb{N}_k = \{0, 1, 2, \ldots, k-1\}$,
$\mathfrak{P}_r(M) = \{E;\ E \subseteq M,\ |E| = r\}$,

$$\operatorname{sgn} x = \begin{cases} 1 & \text{for } x > 0, \\ 0 & \text{for } x = 0, \\ -1 & \text{for } x < 0, \end{cases}$$

$c, c_1, c_2, \ldots$ are positive constants.
Eigenvalues and characteristic polynomial:

$\lambda_1, \ldots, \lambda_{N-1}$ denotes the eigenvalues of the matrix A_N,

$$\lambda_{\max} = \max_{1 \leqslant n \leqslant N-1} |\lambda_n|,$$

$$\chi_N(x) = \det(xI_{N-1} - A_N),$$

$$\hat{\chi}_N(x) = \chi_N(x) \prod_{2 \leqslant j \leqslant N} (x - j)^{-1}.$$

The τ- and σ-coefficients:

$$\tau_{k,N} = \frac{1}{N^{k-1}} \sum_{\substack{1 \leqslant i_1 \leqslant \cdots \leqslant i_k = N \\ i_1 \mid \cdots \mid i_k}} i_1 \cdots i_{k-1} \quad (k \geqslant 1), \qquad \tau_{0,N} = 0,$$

$$\sigma_{k,N} = \frac{1}{N^k} \sum_{\substack{2 \leqslant i_1 \leqslant \cdots \leqslant i_k \leqslant N \\ i_1 \mid \cdots \mid i_k}} i_1 \cdots i_{k-1} \quad (k \geqslant 1), \qquad \sigma_{0,N} = 1,$$

$$S_{k,N} = N^k \sigma_{k,N},$$

$$\tau(N) = \lim_{k \to \infty} \tau_{k,N},$$

$$\sigma_{\infty,N} = \lim_{k \to \infty} \sigma_{k,N} \quad \left(= \frac{N-1}{N^2} \tau(N) \right),$$

$$\sigma_k = \lim_{N \to \infty} \sigma_{k,N} \quad \left(= \frac{1}{k} \prod_{s=2}^{k} \zeta(s) \right).$$

The coefficients g:

$$g_{k,N} = \frac{1}{N^k}\left(\sum_{n=2}^{N} n^k - \sum_{n=1}^{N-1} \lambda_n^k\right),$$

$$g_k = \lim_{N\to\infty} g_{k,N},$$

$$g_{k,N}^* = 1 - g_{k,N}.$$

Partitions:

$\mathscr{P} = \{\rho = (\rho_j)_{j\geqslant 1} \in \mathbb{N}_0^{\infty};\ \Sigma_{j\geqslant 1}\rho_j j < \infty\}$ is the set of all partitions of natural numbers $\geqslant 0$,

$\rho^* = \Sigma_{j\geqslant 1}\rho_j j$ denotes the number which is partitioned by ρ,

$\bar{\rho} = \Sigma_{j\geqslant 1}\rho_j$ is the number of summands of the partition ρ,

$\omega(\rho) = \Pi_{j\geqslant 1}(\rho_j!)$ denotes the weight of ρ,

$\mathscr{P}_r = \{\rho \in \mathscr{P};\ \rho^* = r\}$ is the set of all partitions of r,

$\mathscr{P}(k) = \{\rho \in \mathscr{P};\ \bar{\rho} + \rho^* \leqslant k\}$.

We need several representations and estimates for the τ- and σ-coefficients. The proofs can be found in the Appendix.

LEMMA 1.

(1) $\tau_{k,N}$ *is multiplicative in* N *for every* $k \geqslant 1$.

(2) $(\tau_{k,N})_{k\geqslant 0}$ *increases monotonically.*

(3) $\tau_{k,p^s} = \tau_{k,p^{s-1}} + \dfrac{1}{p^s}\tau_{k-1,p^s} = \displaystyle\prod_{r=1}^{s} \frac{1-p^{1-k-r}}{1-p^{-r}}$.

(4) $\tau(N) = \displaystyle\prod_{p^s \| N}\prod_{r=1}^{s}(1-p^{-r})^{-1} = O(\log_2 N)$.

(5) $\tau_{k,N} = \tau(N) + O\left(\dfrac{\log_2 N}{2^k}\right)$.

LEMMA 2.

(1) $(\sigma_{k,N})_{k\geqslant 0}$ *decreases monotonically.*

(2) $\sigma_{k,N} = \sigma_k + O\left(\dfrac{\log^{2/3} N}{N}\right)$ *and* $\sigma_k = \dfrac{1}{k}\prod_{s=2}^{k} \zeta(s)$.

The O-constant is independent of k.

(3) $\sigma_k = \dfrac{\zeta_\infty}{k} + O\left(\dfrac{1}{k}2^{-k}\right)$.

(4) $\sigma_{\infty,N} = \dfrac{N-1}{N^2}\tau(N)$.

(5) $$\sigma_{k,N} = \sigma_{\infty,N} + O\left(\left(1-\frac{1}{N}\right)^k\left(\frac{1}{k} + \frac{\log^{2/3} N}{N}\right)\right)$$
$$= \sigma_{\infty,N} + O\left(\frac{\log_2 N}{k}\right).$$

(6) $$\sigma_{k,N} - \sigma_{k+1,N} = \frac{1}{k}\left(1-\frac{1}{N}\right)^k \sigma_{k+1,N} + O\left\{\frac{\log^{2/3} N}{kN}\left(1-\frac{1}{N}\right)^k\left(1+\frac{k^2}{N^2}\right) + \frac{1}{2^k}(\log N)(\log_2 N)\right\}.$$

1. DISTRIBUTION OF EIGENVALUES

THEOREM 2. *The eigenvalues* $\lambda_1, \ldots, \lambda_{N-1}$ *of* A_N *lie in the closed disk* $K_{N-1-1/N}(1)$ *of radius* $N-1-1/N$ *and centre* 1. *In particular*

$$|\lambda_n| \leqslant N - \frac{1}{N} \qquad (1 \leqslant n \leqslant N-1).$$

Proof. Conjugation with the matrix

$$H = \begin{pmatrix} -1 & 1 & & & & \\ -1 & 0 & 1 & & & \\ -1 & & 0 & 1 & & \\ -1 & & & 0 & \ddots & \\ \vdots & & & & \ddots & 1 \\ -1 & & & & & 0 \end{pmatrix}$$

gives

$$B_N = H^{-1}A_N H = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & \cdots & 1 \\ r_2 & 2 & & 2 & & 2 & & \\ r_3 & & 3 & & & 3 & & \\ r_4 & & & 4 & & & & \\ r_5 & & & & 5 & & & \\ r_6 & & & & & 6 & & \\ \vdots & & & & & & \ddots & \\ r_{N-1} & & & & & & & N-1 \end{pmatrix},$$

where $r_m = N - m[N/m]$ denotes the smallest nonnegative remainder of N modulo m. We conjugate once more, multiplying the first row of B_N by $N/(N-1)$ and the first column by $(N-1)/N$, and we find that all eigenvalues of A_N lie in the union of the $N-1$ disks:

$K_{\rho_1}(1)$ with

$$\rho_1 = (N-2)\frac{N}{N-1} < N - 1 - \frac{1}{N} =: \rho_1',$$

$K_{\rho_m}(m)$ with

$$\rho_m = r_m \frac{N-1}{N} + m\left(\left[\frac{N-1}{m}\right] - 1\right) \leqslant N - m - \frac{1}{N} =: \rho_m', \qquad 2 \leqslant m \leqslant N-1.$$

Now

$$K_{\rho_m'}(m) \subseteq K_{N-1-1/N}(1) \subseteq K_{N-1/N}(0) \qquad (1 \leqslant m \leqslant N-1)$$

proves the theorem. ∎

THEOREM 3.

(1) $\chi_N(x) = (x-N)\chi_{N-1}(x) + h_N(x)$,

(2) $\chi_N(x) = \Sigma_{1 \leqslant m \leqslant N} h_m(x) \Pi_{m < j \leqslant N}(x-j)$ *for the characteristic polynomial* $\chi_N(x)$ *of* A_N, *where* $\chi_0(x) := 0$, $\chi_1(x) := 1$, *and*

$$h_N(x) := \begin{cases} 1 & (N=1), \\ \displaystyle\sum_{1 \leqslant k < N,\, k \mid N} k\, h_k(x) \prod_{k < j < N} (x-j) & (N>1), \end{cases}$$

(3) $\hat{\chi}_N(x) = \Sigma_{1 \leqslant m \leqslant N} \hat{h}_m(x)$,

(4) $\hat{\chi}_N(x) = \sum_{i \geqslant 0} \sum_{\substack{1 = k_{i+1} < k_i < \cdots < k_1 \leqslant N \\ k_{i+1} \mid k_i \mid \cdots \mid k_1}} \prod_{1 \leqslant j \leqslant i} \frac{k_{j+1}}{x - k_j}$ *for* $\hat{\chi}_N(x) = \chi_N(x) \prod_{2 \leqslant j \leqslant N} (x-j)^{-1}$, $\hat{h}_N(x) = h_N(x) \prod_{2 \leqslant j \leqslant N} (x-j)^{-1}$.

Proof. We subtract the first column from all other columns of the matrix $xI_{N-1} - A_N$ and obtain

$$\begin{pmatrix} x-1 & 2-x & -x & 2-x & -x & 2-x & -x & \cdots & \\ 1 & x-3 & 0 & 0 & -3 & 0 & 0 & & \\ 1 & 0 & x-4 & 0 & 0 & 0 & -4 & & \\ 1 & 0 & 0 & x-5 & 0 & 0 & 0 & & \\ 1 & 0 & 0 & 0 & x-6 & 0 & 0 & & \\ 1 & 0 & 0 & 0 & 0 & x-7 & 0 & & \\ 1 & 0 & 0 & 0 & 0 & 0 & x-8 & & \\ \vdots & & & & & & & \ddots & \\ 1 & & & & & & & & x-N \end{pmatrix} \tag{1.1}$$

and

$$\chi_N(x) = (x-N)\chi_{N-1}(x) + (-1)^N \det H_N,$$

where H_N results from the matrix in (1.1) by deleting the first column and the last row. Now the expansion of $\det H_N$ by its last column is

$$\det H_N = (-1)^N (x + 2\delta_{2 \mid N} - 2) \prod_{3 \leqslant j \leqslant N-1} (x-j)$$

$$+ \sum_{3 \leqslant k < N,\ k \mid N} (-1)^{N+k} k (\det H_k) \prod_{k < j < N} (x-j).$$

With $-\det H_1 := 1 =: \det H_2$ we obtain

$$(-1)^N \det H_N = \sum_{1 \leqslant k < N,\ k \mid N} k(-1)^k (\det H_k) \prod_{k < j < N} (x-j),$$

and $h_N(x) := (-1)^N \det H_N$ gives Equation (1).

Division of (1) by $\prod_{2 \leqslant j \leqslant N}(x-j)$ gives

$$\hat{\chi}_N(x) = \hat{\chi}_{N-1}(x) + \hat{h}_N(x),$$

and summation over $1,2,\ldots,N$ leads to Equations (3) and (2).

Now we prove for every $r \geqslant 0$

$$\hat{\chi}_N(x) = \sum_{i=0}^{r-1} \sum_{\substack{1 = k_{i+1} < k_i < \cdots < k_1 \leqslant N \\ k_{i+1} \mid k_i \mid \cdots \mid k_1}} \prod_{j=1}^{i} \frac{k_{j+1}}{x-k_j} + \sum_{\substack{1 \leqslant k_{r+1} < \cdots < k_1 \leqslant N \\ k_{r+1} \mid \cdots \mid k_1}} \hat{h}_{k_{r+1}}(x) \prod_{j=1}^{r} \frac{k_{j+1}}{x-k_j}. \tag{1.2}$$

We proceed by induction: For $r=0$ this is Equation (3), and

$$\begin{aligned}
\hat{\chi}_N(x) &= \sum_{i=0}^{r-1} \sum_{\substack{1 = k_{i+1} < \cdots < k_1 \leqslant N \\ k_{i+1} \mid \cdots \mid k_1}} \prod_{j=1}^{i} \frac{k_{j+1}}{x-k_j} \\
&\quad + \sum_{\substack{1 = k_{r+1} < \cdots < k_1 \leqslant N \\ k_{r+1} \mid \cdots \mid k_1}} \prod_{j=1}^{r} \frac{k_{j+1}}{x-k_j} \\
&\quad + \sum_{\substack{1 < k_{r+1} < \cdots < k_1 \leqslant N \\ k_{r+1} \mid \cdots \mid k_1}} \hat{h}_{k_{r+1}}(x) \prod_{j=1}^{r} \frac{k_{j+1}}{x-k_j} \\
&= \sum_{i=0}^{r} \sum_{\substack{1 = k_{i+1} < \cdots < k_1 \leqslant N \\ k_{i+1} \mid \cdots \mid k_1}} \prod_{j=1}^{i} \frac{k_{j+1}}{x-k_j} \\
&\quad + \sum_{\substack{1 < k_{r+1} < \cdots < k_1 \leqslant N \\ k_{r+1} \mid \cdots \mid k_1}} \left\{ \frac{1}{x-k_{r+1}} \sum_{\substack{1 \leqslant k_{r+2} < k_{r+1} \\ k_{r+2} \mid k_{r+1}}} k_{r+2} \hat{h}_{k_{r+2}}(x) \right\} \\
&\quad \times \prod_{j=1}^{r} \frac{k_{j+1}}{x-k_j},
\end{aligned}$$

for the definition of $h_N(x)$ shows for $N \geqslant 2$

$$\hat{h}_N(x) = \frac{1}{x-N} \sum_{1 \leqslant k < N,\ k \mid N} k\,\hat{h}_k(x).$$

This is (1.2) for $r+1$ instead of r. The second sum in (1.2) disappears for large r, and so we arrive at Equation (4). ∎

THEOREM 4. *For* $1 \leqslant n \leqslant N$,

$$\frac{(-1)^N}{(N-n)!}\chi_N(n) = (-1)^{n-1} n! \frac{\tau(n)}{n^2} \frac{(-1)^{[N/n]}}{([N/n]-1)!}\chi_{[N/n]}(1),$$

where

$$\tau(n) = \prod_{p^s \| n} \prod_{1 \leqslant r \leqslant s} (1 - p^{-r})^{-1}.$$

Proof. The assertion is clear for $n = 1$, and we assume $2 \leqslant n \leqslant N$. The equation

$$\chi_N(x) = \sum_{i \geqslant 0} \sum_{\substack{1 = k_{i+1} < \cdots < k_1 \leqslant N \\ k_{i+1} \mid \cdots \mid k_1}} \left(\prod_{j=1}^{i} \frac{k_{j+1}}{x - k_j} \right) \prod_{m=2}^{N} (x - m)$$

in Theorem 3(4) shows that at most those summands on the right in which one of the divisors $k_1, \ldots, k_i$ equals n contribute something to $\chi_N(n)$. Therefore we obtain

$$\chi_N(n) = \sum_{r \geqslant 1} \sum_{i \geqslant 0} \sum_{\substack{1 = k_{i+1} < \cdots < k_r = n \\ n < k_{r-1} < \cdots < k_1 \leqslant N \\ k_{i+1} \mid \cdots \mid k_r \mid \cdots \mid k_1}} \frac{\prod_{j=1}^{i} k_{j+1}}{\prod_{\substack{j=1 \\ j \neq r}}^{i} (n - k_j)}$$

$$\times (n-2)!(N-n)!(-1)^{N-n}$$

and

$$\frac{(-1)^N}{(N-n)!}\chi_N(n)\{(-1)^n(n-2)!\}^{-1}$$

$$= \sum_{r\geqslant 1}\sum_{\substack{n=k_r<\cdots<k_1\leqslant N\\ k_r\mid\cdots\mid k_1}}\left(\prod_{j=1}^{r-1}\frac{k_{j+1}}{n-k_j}\right)$$

$$\times\sum_{i\geqslant 0}\sum_{\substack{1=k_{i+1}<\cdots<k_{r+1}<n\\ k_{i+1}\mid\cdots\mid k_{r+1}\mid n}}\prod_{j=r+1}^{i}\frac{k_j}{n-k_j}$$

$$= \left\{\sum_{r\geqslant 1}\sum_{\substack{1=k'_r<\cdots<k'_1\leqslant N/n\\ k'_r\mid\cdots\mid k'_1}}\prod_{j=1}^{r-1}\frac{k'_{j+1}}{1-k'_j}\right\}$$

$$\times\left\{\sum_{i\geqslant 1}\sum_{\substack{1=k_i<\cdots<k_1<n\\ k_i\mid\cdots\mid k_1\mid n}}\prod_{j=1}^{i-1}\frac{k_j}{n-k_j}\right\}.$$

Theorem 3(4) shows that the first factor in this product is

$$\hat{\chi}_{[N/n]}(1)=\frac{(-1)^{[N/n]-1}}{([N/n]-1)!}\chi_{[N/n]}(1),$$

and it follows that

$$\frac{(-1)^N}{(N-n)!}\chi_N(n)=(-1)^{n-1}(n-2)!$$

$$\times\frac{(-1)^{[N/n]}}{([N/n]-1)!}\chi_{[N/n]}(1)\gamma(n), \qquad (1.3)$$

where

$$\gamma(n)=\sum_{i\geqslant 1}\sum_{\substack{1=k_i<\cdots<k_1<n\\ k_i\mid\cdots\mid k_1\mid n}}\prod_{j=1}^{i-1}\frac{k_j}{n-k_j}.$$

To prove $\gamma(n) = \{(n-1)/n\}\tau(n)$ we use the result

$$\chi_N(N) = \frac{(N-1)!}{N}\tau(N)$$

from the Corollary of Theorem 11 in Section 3: (1.3) says for $n = N$

$$\chi_N(N) = (N-2)!\chi_1(1)\gamma(N),$$

and hence the Corollary gives $\{(N-1)/N\}\tau(N) = \gamma(N)$ because of $\chi_1(1) = 1$. ∎

THEOREM 5.

(1) *All natural numbers n in the interval* $]N/3, N/2]$ *are eigenvalues of* A_N.

(2) *If* $1 \leqslant n \leqslant N-1$ *and if n is not an element of the sequence* $([N/m])_{2 \leqslant m \leqslant N}$, *then the interval* $[n, n+1[$ *contains an eigenvalue of* A_N.

(3) *There are eigenvalues of* A_N *in at least* $N - 2[\sqrt{N}]$ *different intervals* $[n, n+1[$, $1 \leqslant n \leqslant N-1$.

Proof. From Theorem 4 follows

$$\operatorname{sgn}\chi_N^*(n) = (-1)^{n-1}\operatorname{sgn}\chi_{[N/n]}^*(1), \tag{1.4}$$

where $\chi_N^*(x) := (-1)^N\chi_N(x)$.

(1): If $N/3 < n \leqslant N/2$, then $[N/2] = 2$ and $\chi_2(1) = 0$. Thus the assertion derives immediately from (1.4).

(2): We assume n not to be an element of $([N/m])_{2 \leqslant m \leqslant N}$, i.e. $[N/(n+1)] = [N/n]$. If then $\chi_N^*(n) \neq 0$, (1.4) for n and $n+1$ shows

$$\begin{aligned} 0 \neq \operatorname{sgn}\chi_N^*(n) &= (-1)^{n-1}\operatorname{sgn}\chi_{[N/n]}^*(1) \\ &= (-1)^{n-1}\operatorname{sgn}\chi_{[N/(n+1)]}^*(1) \\ &= -\operatorname{sgn}\chi_N^*(n+1); \end{aligned}$$

hence there must be a zero of $\chi_N^*(x)$ in $]n, n+1[$.

(3): Let α_N count the different numbers in the sequence $([N/m])_{1 \leqslant m \leqslant N}$. In regard to (2) it is sufficient to prove

$$\alpha_N \leqslant 2[\sqrt{N}].$$

Now, r occurs in $([N/m])_{1 \leqslant m \leqslant N}$ if and only if the interval $I_r =]N/(r+1), N/r]$ contains a natural number m. In $\bigcup_{1 \leqslant r \leqslant \sqrt{N}} I_r$ lie all m such that $N/([\sqrt{N}]+1) < m \leqslant N$. Hence there remain only $[N/([\sqrt{N}]+1)]$ numbers m which lie in intervals I_r for $r > [\sqrt{N}]$, and thus

$$\alpha_N \leqslant [\sqrt{N}] + \left[\frac{N}{[\sqrt{N}]+1}\right] \leqslant 2[\sqrt{N}]. \qquad \blacksquare$$

2. POWER SUMS

In Section 1 we have seen that almost all eigenvalues $\lambda_1, \ldots, \lambda_{N-1}$ of A_N satisfy the estimate $n \leqslant \lambda_n < n+1$. This suggests the comparison of the power sums of the eigenvalues and those of the numbers n.

THEOREM 6. *For $k \geqslant 0$, $N \geqslant 2$,*

$$\sum_{n=1}^{N-1} \lambda_n^k = \sum_{n=2}^{N} n^k - g_{k,N} N^k, \qquad (*)$$

where

$$g_{k,N} = k\sigma_{k,N} - \sum_{i=1}^{k-1} \sigma_{k-i,N} g_{i,N}.$$

REMARK. Equation (*) and Theorem 2 immediately imply

$$\lim_{k \to \infty} g_{k,N} = 1.$$

More interesting however seems to be the course of values of the finite sequence $(g_{k,N})_{0 \leqslant k \leqslant cN \log^2 N}$.

Proof. Every set $E \in \mathfrak{P}_r(\mathbb{N}_k)$, $1 \leqslant r < k$, i.e. every subset of $\{0,1,2,\ldots, k-1\}$ containing r elements, can be uniquely divided into sequences of maximal length of successive numbers modulo k,

$$E = \bigcup_{v=1}^{t} \{m_v, m_v+1, \ldots, m_v + r_v - 1\}, \tag{2.1}$$

so that t is as small as possible, and

$$\sum_{v=1}^{t} r_v = r, \qquad r_v \geqslant 1 \quad \text{for } 1 \leqslant v \leqslant t.$$

Let ρ_E denote this partition of r:

$$\rho_E = (\rho_j)_{j \geqslant 1} \in \mathscr{P}_r \text{ with } \rho_j = |\{v;\ r_v = j,\ 1 \leqslant v \leqslant t\}|.$$

This defines a mapping

$$\Phi_{r,k}: \mathfrak{P}_r(\mathbb{N}_k) \to \mathscr{P}_r, \qquad \Phi_{r,k}(E) = \rho_E,$$

and for $\rho \in \mathscr{P}_r$

$$\phi_{r,k}(\rho) := |\{E \in \mathfrak{P}_r(\mathbb{N}_k);\ \Phi_{r,k}(E) = \rho\}|$$

is the number of sets E with image ρ. We need three propositions.

PROPOSITION 1. *For $1 \leqslant r < k$ we have*

$$\phi_{r,k}(\rho) = \begin{cases} \dfrac{k}{\omega(\rho)} \dfrac{(k-r-1)!}{(k-r-\bar{\rho})!} & \text{if } \ r + \bar{\rho} \leqslant k, \\ 0 & \text{if } \ r + \bar{\rho} > k. \end{cases}$$

Proof. We represent $E \in \mathfrak{P}_r(\mathbb{N}_k)$ in the standardized form (2.1) and observe that then

$$E^+ := \bigcup_{v=1}^{t} \{m_v, m_v+1, \ldots, m_v + r_v\}$$

is also a disjoint union of t sequences of successive numbers modulo k, and

$$k \geqslant |E^+| = \sum_{v=1}^{t} (r_v + 1) = r + \bar{\rho}_E.$$

Now we identify the sequences $\{m_v, m_v + 1, \ldots, m_v + r_v\}$ with the arcs

$$\left\{ e^{2\pi i x/k};\ m_v \leqslant x \leqslant m_v + r_v \right\}$$

of the unit circle S^1. If $\rho = (\rho_j)_{j \geqslant 1}$ is in $\mathscr{P}_r$ and $\Phi_{r,k}(E) = \rho$, then exactly $\bar{\rho}$ such arcs belong to ρ, via the standardized representation of E in (2.1), and these arcs are disjoint in pairs. We remove them from S^1, so that we get $\bar{\rho}$ gaps, i.e. open arcs. The length of such an open arc will be measured by the number of kth roots of unity lying on it.

If we foresee for a moment the lengths of these gaps, there are

$$k \frac{(\bar{\rho} - 1)!}{\omega(\rho)} \tag{2.2}$$

different possibilities to place the arcs, which are given by ρ, on S^1. But in every case we can choose the $\bar{\rho}$ gaps in $\gamma(L, \bar{\rho})$ different ways, where $L = k - r - \bar{\rho}$ is the number of the kth roots of unity lying in these gaps and $\gamma(L, \bar{\rho})$ denotes the number of sequences $(a_j)_{1 \leqslant j \leqslant \bar{\rho}}$ in $\mathbb{N}_0$ such that $a_1 + \cdots + a_{\bar{\rho}} = L$. These sequences determine the lengths of the gaps between the removed arcs.

Induction on m shows

$$\gamma(n, m+1) = \binom{n+m}{m}, \tag{2.3}$$

for

$$1 = \binom{n}{0}$$

is the number of sequences (a_1) such that $a_1 = n$, and the number of sequences $(a_j)_{1 \leqslant j \leqslant m+1}$ with sum n is

$$\sum_{j=0}^{n} \gamma(j, m) = \sum_{j=0}^{n} \binom{j+m-1}{m-1} = \binom{n+m}{m},$$

because a_{m+1} can take the values $0, 1, \ldots, n$.

Finally (2.2) and (2.3) imply

$$\phi_{r,k}(\rho) = k\frac{(\bar{\rho}-1)!}{\omega(\rho)}\gamma(k-r-\bar{\rho},\bar{\rho})$$
$$= \frac{k}{\omega(\rho)}\frac{(k-r-1)!}{(k-r-\bar{\rho})!}.$$

■

PROPOSITION 2. *For $k \geqslant 1$ and $N \geqslant 2$ we have*

$$\sum_{n=1}^{N-1}\lambda_n^k = \sum_{n=2}^{N} n^k + (N-1)^k$$
$$\times \sum_{\rho\in\mathscr{P}(k)} (-1)^{k-\rho^*}\frac{k}{\omega(\rho)}\frac{(k-\rho^*-1)!}{(k-\rho^*-\bar{\rho})!}\prod_{j\geqslant 1}\left(\frac{S_{j+1,N}}{(N-1)^{j+1}}\right)^{\rho_j}.$$

Proof. The definition of A_N shows

$$\sum_{n=1}^{N-1}\lambda_n^k = \operatorname{tr}A_N^k = \sum_{2\leqslant i_0,\ldots,i_{k-1}\leqslant N}\prod_{j \bmod k}\left(i_j\delta_{i_j\mid i_{j+1}}-1\right).$$

Now we arrange the k-ple products according to the number r $(0 \leqslant r \leqslant k)$ of their δ-factors:

$$\operatorname{tr}A_N^k = \sum_{2\leqslant i_0,\ldots,i_{k-1}\leqslant N}(-1)^k + \sum_{r=1}^{k-1}(-1)^{k-r}\Delta_r$$
$$+ \sum_{\substack{2\leqslant i_0\leqslant\cdots\leqslant i_{k-1}\leqslant N\\ i_0\mid\cdots\mid i_{k-1}\mid i_0}} i_0\cdots i_{k-1}$$
$$= (-1)^k(N-1)^k + \sum_{r=1}^{k-1}(-1)^{k-r}\Delta_r + \sum_{2\leqslant n\leqslant N} n^k \qquad (2.4)$$

and

$$\Delta_r = \sum_{2\leqslant i_0,\ldots,i_{k-1}\leqslant N}\sum_{E\in\mathfrak{P}_r(\mathbb{N}_k)}\prod_{m\in E}\left(i_m\delta_{i_m\mid i_{m+1}}\right)$$
$$= \sum_{E\in\mathfrak{P}_r(\mathbb{N}_k)}\sum_{\substack{2\leqslant i_0,\ldots,i_{k-1}\leqslant N\\ i_m\mid i_{m+1}\text{ for }m\in E}}\prod_{m\in E} i_m.$$

$E \in \mathfrak{P}_r(\mathbb{N}_k)$ with $\Phi_{r,k}(E) = \rho$, i.e. of the form (2.1) with

$$\sum_{v=1}^{t} r_v = \rho^*, \qquad t = \bar{\rho},$$

supplies to Δ_r the contribution

$$\sum_{\substack{2 \leqslant i_0, \ldots, i_{k-1} \leqslant N \\ i_{m_v} \mid i_{m_v+1} \mid \cdots \mid i_{m_v+r_v} \text{ for } 1 \leqslant v \leqslant t}} \prod_{v=1}^{t} \left(i_{m_v} i_{m_v+1} \cdots i_{m_v+r_v-1} \right)$$

$$= (N-1)^{k-\rho^*-\bar{\rho}} \prod_{v=1}^{t} \left\{ \sum_{\substack{2 \leqslant i_{m_v} \leqslant \cdots \leqslant i_{m_v+r_v} \leqslant N \\ i_{m_v} \mid \cdots \mid i_{m_v+r_v}}} i_{m_v} \cdots i_{m_v+r_v-1} \right\}$$

$$= (N-1)^{k-\rho^*-\bar{\rho}} \prod_{v=1}^{t} S_{r_v+1,N}$$

$$= (N-1)^{k-\rho^*-\bar{\rho}} \prod_{j \geqslant 1} S_{j+1,N}^{\rho_j},$$

for $r_1, \ldots, r_t$ are the summands of the partition $\rho = (\rho_j)_{j \geqslant 1}$ of r. In view of Proposition 1 we therefore obtain

$$\Delta_r = \sum_{\rho \in \mathscr{P}_r} \phi_{r,k}(\rho)(N-1)^{k-\rho^*-\bar{\rho}} \prod_{j \geqslant 1} S_{j+1,N}^{\rho_j}$$

$$= \sum_{\substack{\rho \in \mathscr{P}_r \\ \rho^* + \bar{\rho} \leqslant k}} \frac{k}{\omega(\rho)} \frac{(k-\rho^*-1)!}{(k-\rho^*-\bar{\rho})!} (N-1)^k \prod_{j \geqslant 1} \left(\frac{S_{j+1,N}}{(N-1)^{j+1}} \right)^{\rho_j},$$

and this, combined with (2.4), proves Proposition 2. ∎

Now we consider the polynomials

$$F_k = F_k(x_1, \ldots, x_{k-1})$$

$$= \sum_{\rho \in \mathscr{P}(k)} (-1)^{k-1-\rho^*} \frac{k}{\omega(\rho)} \frac{(k-\rho^*-1)!}{(k-\rho^*-\bar{\rho})!} X^{\rho},$$

where X^{ρ} is used as an abbreviation for $\prod_{j \geqslant 1} x_j^{\rho_j}$. Clearly, $F_1 = 1$, and

PROPOSITION 3. $F_k = kx_{k-1} - \sum_{i=1}^{k-2} x_{k-1-i}F_i - F_{k-1}$ *for* $k \geqslant 2$.

Proof. First we observe that all monomials X^ρ which occur on the right of the equation have exponents $\rho \in \mathscr{P}(k)$. For x_{k-1} is X^ρ with $\rho = (\rho_j)_{j\geqslant 1} = (\delta_{j,k-1})_{j\geqslant 1}$; therefore $\bar{\rho} + \rho^* = 1 + (k-1) = k$. And $x_{k-1-i}F_i$ consists of monomials of the form $X^\rho = x_{k-1-i}X^\tau$ such that $\tau = (\tau_j)_{j\geqslant 1} \in \mathscr{P}(i)$, so we get $\rho = (\tau_j + \delta_{j,k-1-i})_{j\geqslant 1}$ and hence

$$\begin{aligned}\bar{\rho} + \rho^* &= (\bar{\tau} + 1) + (\tau^* + (k-1-i)) = \bar{\tau} + \tau^* + k - i \\ &\leqslant i + k - i = k.\end{aligned}$$

Now we take $\rho \in \mathscr{P}(k)$,

$$\rho^* = \sum_{v=1}^{t} \rho_{j_v} j_v, \qquad 1 \leqslant j_1 < \cdots < j_t \leqslant k-1, \quad \rho_{j_1} \geqslant 1, \ldots, \rho_{j_t} \geqslant 1,$$

and compare the coefficients of X^ρ on both sides of the equation in the proposition. We distinguish three cases:

(1) $\bar{\rho} + \rho^* = k$, $\rho_{k-1} = 1$,
(2) $\bar{\rho} + \rho^* = k$, $\rho_{k-1} \neq 1$,
(3) $\bar{\rho} + \rho^* < k$.

(1): $\rho_{k-1} = 1$ implies $\bar{\rho} = 1$, $\rho^* = k-1$; thus $X^\rho = x_{k-1}$. The coefficient of x_{k-1} in F_k is k, and x_{k-1} does not appear either in $\sum_{i=1}^{k-2} x_{k-1-i}F_i$ or in F_{k-1}.

(2): $\rho_{k-1} \neq 1$ implies $\rho_{k-1} = 0$; thus $1 \leqslant j_1 < \cdots < j_t \leqslant k-2$. X^ρ does not appear in F_{k-1} because of $\bar{\rho} + \rho^* = k$, but it appears t times in $\sum_{i=1}^{k-2} x_i F_{k-1-i}$, via the factorizations

$$X^\rho = x_{j_v}\left(x_{j_1}^{\rho_{j_1}} \cdots x_{j_v}^{\rho_{j_v}-1} \cdots x_{j_t}^{\rho_{j_t}}\right) =: x_{j_v} X^{\rho_{(v)}}, \qquad 1 \leqslant v \leqslant t,$$

i.e. for $i = j_1, \ldots, j_t$. So we find as coefficient of X^ρ on the right-hand side

$$\begin{aligned}&-\sum_{v=1}^{t} (-1)^{(k-1-j_v)-1-\rho^*_{(v)}} \frac{k-1-j_v}{\omega(\rho_{(v)})} \frac{\{(k-1-j_v)-\rho^*_{(v)}-1\}!}{0!} \\ &= \sum_{v=1}^{t} (-1)^{k-1-\rho^*} \frac{\rho_{j_v}(k-1-j_v)}{\omega(\rho)} (k-\rho^*-2)! \\ &= (-1)^{k-1-\rho^*} \frac{(k-\rho^*-2)!}{\omega(\rho)} \sum_{v=1}^{t} \rho_{j_v}(k-1-j_v) \\ &= (-1)^{k-1-\rho^*} \frac{k}{\omega(\rho)} (k-\rho^*-1)!, \end{aligned} \tag{2.5}$$

for $\bar{\rho} + \rho^* = k$ shows

$$\sum_{v=1}^{t} \rho_{j_v}(k-1-j_v) = (k-1)\bar{\rho} - \rho^* = k\bar{\rho} - k = k(k-\rho^*-1),$$

and (2.5) is also the coefficient of X^ρ in F_k.

(3): X^ρ appears in F_{k-1} with the coefficient

$$(-1)^{k-2-\rho^*} \frac{k-1}{\omega(\rho)} \frac{(k-\rho^*-2)!}{(k-1-\rho^*-\bar{\rho})!}. \tag{2.6}$$

As in case (2), X^ρ occurs in $\sum_{i=1}^{k-2} x_i F_{k-1-i}$ exactly t times, here with the coefficient

$$\sum_{v=1}^{t} (-1)^{(k-1-j_v)-1-\rho^*_{(v)}} \frac{k-1-j_v}{\omega(\rho_{(v)})} \frac{\{(k-1-j_v)-\rho^*_{(v)}-1\}!}{\{(k-1-j_v)-\rho^*_{(v)}-\overline{\rho_{(v)}}\}!}$$

$$= \sum_{v=1}^{t} (-1)^{k-\rho^*} \frac{\rho_{j_v}(k-1-j_v)}{\omega(\rho)} \frac{(k-\rho^*-2)!}{(k-\rho^*-\bar{\rho})!}$$

$$= (-1)^{k-\rho^*} \frac{(k-\rho^*-2)!}{\omega(\rho)(k-\rho^*-\bar{\rho})!} \sum_{v=1}^{t} \rho_{j_v}(k-1-j_v)$$

$$= (-1)^{k-\rho^*} \frac{(k-\rho^*-2)!}{\omega(\rho)(k-\rho^*-\bar{\rho})!} \{(k-1)\bar{\rho} - \rho^*\}.$$

This and (2.6) show that the coefficient of X^ρ on the right of the equation in Proposition 3 is

$$-(-1)^{k-\rho^*} \frac{(k-\rho^*-2)!}{(k-\rho^*-\bar{\rho})!} \frac{1}{\omega(\rho)} \{(k-1)\bar{\rho} - \rho^* + (k-1)(k-\rho^*-\bar{\rho})\}$$

$$= (-1)^{k-1-\rho^*} \frac{(k-\rho^*-2)!}{(k-\rho^*-\bar{\rho})!} \frac{1}{\omega(\rho)} k(k-\rho^*-1),$$

which is also the coefficient of X^ρ in F_k. ■

Now we can finish the proof of Theorem 6: From Proposition 2 and the definition of F_k it follows that

$$\sum_{n=1}^{N-1} \lambda_n^k = \sum_{n=2}^{N} n^k - (N-1)^k F_k\left(\frac{S_{2,N}}{(N-1)^2}, \ldots, \frac{S_{k,N}}{(N-1)^k}\right)$$

$$= \sum_{n=2}^{N} n^k - g_{k,N} N^k$$

with $g_{0,N} = 0$ and

$$g_{k,N} = \left(1 - \frac{1}{N}\right)^k F_k\left(\frac{S_{2,N}}{(N-1)^2}, \ldots, \frac{S_{k,N}}{(N-1)^k}\right), \qquad k \geqslant 1.$$

Therefore we have

$$g_{1,N} = \left(1 - \frac{1}{N}\right) F_1 = 1 - \frac{1}{N} = \sigma_{1,N},$$

and Proposition 3 shows for $k \geqslant 2$

$$g_{k,N} = \frac{(N-1)^k}{N^k}\left\{k\frac{S_{k,N}}{(N-1)^k} - \sum_{i=1}^{k-2} \frac{S_{k-i,N}}{(N-1)^{k-i}} F_i\left(\frac{S_{2,N}}{(N-1)^2}, \ldots\right) - F_{k-1}\left(\frac{S_{2,N}}{(N-1)^2}, \ldots\right)\right\}$$

$$= k\frac{S_{k,N}}{N^k} - \sum_{i=1}^{k-2} \frac{S_{k-i,N}}{N^{k-i}} \frac{(N-1)^i}{N^i} F_i\left(\frac{S_{2,N}}{(N-1)^2}, \ldots\right)$$

$$- \left(1 - \frac{1}{N}\right)\frac{(N-1)^{k-1}}{N^{k-1}} F_{k-1}\left(\frac{S_{2,N}}{(N-1)^2}, \ldots\right)$$

$$= k\sigma_{k,N} - \sum_{i=1}^{k-1} \sigma_{k-i,N} g_{i,N}. \qquad \blacksquare$$

To investigate the coefficients $g_{k,N}$ we first simplify them: With

$$g_k := \lim_{N \to \infty} g_{k,N}$$

and according to Lemma 2(2), we find, in analogy to the recursion formula of Theorem 6,

$$g_k = k\sigma_k - \sum_{i=1}^{k-1} \sigma_{k-i} g_i \qquad (k \geqslant 0), \tag{2.7}$$

where $\sigma_0 = 1$ and $\sigma_k = (1/k)\prod_{s=2}^{k} \zeta(s)$ for $k \geqslant 1$. The first g_k-values are

$$g_0 = 0, \qquad g_1 = 1, \quad g_2 = \tfrac{1}{6}\pi^2 - 1 = 0.6449\ldots,$$

$$g_3 = 0.5099\ldots, \qquad g_4 = 0.4406\ldots.$$

For in comparison with N small exponents k the g_k's will prove to be fairly good approximations of the $g_{k,N}$'s, but for large k this property disappears slowly: The next theorem implies $\lim_{k \to \infty} g_k = 0$, whereas we have seen that $\lim_{k \to \infty} g_{k,N} = 1$.

THEOREM 7.

$$g_k = \frac{1}{\log k}\{1 + o(1)\} \qquad \textit{for} \quad k \to \infty.$$

Proof. First we observe that for $k \geqslant 2$

$$1 + \frac{1}{2^k} \leqslant \zeta(k) \leqslant 1 + \frac{k+1}{k-1}\frac{1}{2^k}, \tag{2.8}$$

and this implies that

$$(\sigma_k)_{k \geqslant 0} \text{ is decreasing,} \tag{2.9}$$

and further, that for $k \geqslant 1$

$$\frac{\zeta_\infty}{k} - \frac{\zeta_\infty}{2^{k-2}k} \leqslant \sigma_k \leqslant \frac{\zeta_\infty}{k}. \tag{2.10}$$

Taking differences, (2.7) shows

$$g_{k+1} = \sum_{i=1}^{k} (\sigma_{k-i} - \sigma_{k+1-i}) g_i + \{(k+1)\sigma_{k+1} - k\sigma_k\}. \tag{2.11}$$

This and (2.9) give by induction

$$g_k > 0 \qquad (k \geqslant 1), \tag{2.12}$$

for

$$(k+1)\sigma_{k+1} - k\sigma_k > 0. \tag{2.13}$$

Now we prove

$$g_k \leqslant g_2 = \tfrac{1}{6}\pi^2 - 1 < 0.65 \qquad (k \geqslant 2). \tag{2.14}$$

This is clear for $k = 3$ and $k = 4$. For $k \geqslant 4$ we proceed by induction, using (2.11):

$$\begin{aligned} g_{k+1} &\leqslant (\sigma_{k-1} - \sigma_k) + g_2(1 - \sigma_{k-1}) + \zeta_\infty\{\zeta(k+1) - 1\} \\ &\leqslant g_2 + \left\{ \sigma_{k-1} - \sigma_k - \left(\tfrac{1}{6}\pi^2 - 1\right)\sigma_{k-1} \right. \\ &\qquad \left. + \zeta_\infty\{\zeta(k+1) - 1\} \right\}, \end{aligned}$$

and from (2.8) and (2.10) it follows that the value in the curly braces is negative for $k \geqslant 4$.

Now we introduce the power series

$$G(x) := \sum_{k \geqslant 0} g_{k+1} x^k, \qquad S(x) := \sum_{k \geqslant 0} \sigma_k x^k.$$

(2.7) shows

$$G(x)S(x) = \sum_{k \geqslant 0} \left(\sum_{i=1}^{k+1} g_i \sigma_{k+1-i} \right) x^k = S'(x). \tag{2.15}$$

We set

$$S(x)^{-1} =: \sum_{i \geqslant 0} \alpha_i x^i,$$

and equate the coefficients in $G(x) = S'(x)S(x)^{-1}$, which gives

$$g_{k+1} = \sum_{j=0}^{k} (j+1)\sigma_{j+1}\alpha_{k-j}, \tag{2.16}$$

and in view of the estimate of the σ_j's in (2.10),

$$g_{k+1} = \zeta_\infty \sum_{j=0}^{k} \alpha_j + O\left(\sum_{j=0}^{k} \frac{|\alpha_{k-j}|}{2^j} \right). \tag{2.17}$$

For $0 < x < 1$ the estimate (2.10) gives

$$\begin{aligned} S(x) &= \sum_{j \geq 0} \sigma_j x^j = \sum_{j \geq 1} \left\{ \frac{\zeta_\infty}{j} + O\left(\frac{1}{2^j j} \right) \right\} x^j + O(1) \\ &= \zeta_\infty \log \frac{1}{1-x} + O(1), \\ S'(x) &= \sum_{j \geq 1} j\sigma_j x^{j-1} = \zeta_\infty \sum_{j \geq 1} x^{j-1} + O(1) \\ &= \zeta_\infty \frac{1}{1-x} + O(1), \end{aligned}$$

and with (2.15) we obtain for $x \uparrow 1$

$$\sum_{k \geq 0} g_{k+1} x^k = \frac{S'(x)}{S(x)} = \frac{1}{1-x} \left\{ \log^{-1}\left(\frac{1}{1-x} \right) \right\} \{1 + o(1)\}.$$

Because the series on the left has nonnegative coefficients, the Tauberian theorem of Hardy, Littlewood, and Karamata [3, p. 129] gives

$$\sum_{k=1}^{K} g_k = \frac{K}{\log K} \{1 + o(1)\} \qquad \text{for} \quad K \to \infty. \tag{2.18}$$

Differentiation shows, in view of (2.15),

$$\{S(x)^{-1}\}' = -S(x)^{-1} \frac{S'(x)}{S(x)} = -S(x)^{-1} G(x).$$

By equating coefficients, we obtain

$$j\alpha_j = -\sum_{r=0}^{j-1} \alpha_r g_{j-r}, \tag{2.19}$$

and, with the Hölder inequality, for $j \geqslant 1$

$$\alpha_j^2 \leqslant \frac{1}{j^2}\left(\sum_{r=0}^{j-1} \alpha_r^2\right)\left(\sum_{r=1}^{j} g_r^2\right). \tag{2.20}$$

Now we can prove

$$\alpha_j = O(j^{-1/2}). \tag{2.21}$$

From (2.12), (2.14), and (2.18) we obtain for $j \geqslant j_0$

$$\sum_{r=1}^{j} g_r^2 \leqslant 1 + g_2\left(\sum_{r=2}^{j} g_r\right) \leqslant 0.7\frac{j}{\log j}.$$

Next we choose $c \geqslant 1$ so that

$$\alpha_j^2 \leqslant \frac{c^2}{j} \qquad \textit{for} \quad 1 \leqslant j \leqslant j_0 \tag{2.22}$$

and find with (2.20), for $j \geqslant j_0$,

$$\alpha_j^2 \leqslant 0.7\frac{1}{j \log j}\sum_{r=0}^{j-1} \alpha_r^2.$$

If now $j > j_0$, and (2.22) is valid for all indices smaller than j, the last formula implies—because $\alpha_0 = 1$, $\alpha_1 = -1$—

$$\alpha_j^2 \leqslant 0.7\frac{1}{j \log j}\left(2 + c^2 \sum_{r=2}^{j-1} \frac{1}{r}\right)$$

$$\leqslant \frac{c^2}{j}\left(\frac{1.4}{\log j} + 0.7\right) \leqslant \frac{c^2}{j},$$

provided $j_0 \geqslant e^{14/3}$.

(2.21) simplifies (2.17) to

$$g_{k+1} = \zeta_\infty \sum_{j=0}^{k} \alpha_j + O(k^{-1/2}). \tag{2.23}$$

This is now used in the summation of the equations in (2.19):

$$\begin{aligned}\sum_{i=1}^{j} i\alpha_i &= -\sum_{i=1}^{j} g_i \left(\sum_{r=0}^{j-i} \alpha_r \right) \\ &= -\frac{1}{\zeta_\infty} \sum_{i=1}^{j} g_i \left\{ g_{j+1-i} + O\left((j+1-i)^{-1/2} \right) \right\} \\ &= -\frac{1}{\zeta_\infty} \sum_{i=1}^{j} g_i g_{j+1-i} + O\left(\sum_{i=1}^{j} i^{-1/2} \right)\end{aligned}$$

for $0 \leqslant g_i \leqslant 1$. As a consequence we obtain

$$-\zeta_\infty \sum_{i=1}^{j} i\alpha_i = \sum_{i=1}^{j} g_i g_{j+1-i} + O(j^{1/2}). \tag{2.24}$$

Summation of the equations in (2.23) gives

$$\sum_{j=1}^{k} g_j = \zeta_\infty \sum_{j=1}^{k} j\alpha_{k-j} + O(k^{1/2});$$

hence

$$\zeta_\infty k \sum_{j=0}^{k-1} \alpha_j - \sum_{j=1}^{k} g_j = \zeta_\infty \sum_{j=1}^{k-1} j\alpha_j + O(k^{1/2}).$$

In view of (2.23) and (2.24) this means

$$kg_k - \sum_{j=1}^{k} g_j = -\sum_{i=1}^{k-1} g_i g_{k-i} + O(k^{1/2}), \tag{2.25}$$

and the sum on the right is nonnegative. This and (2.18) imply

$$g_k \leqslant \frac{c}{\log(k+1)} \qquad (k \geqslant 1).$$

But then

$$\sum_{i=1}^{k-1} g_i g_{k-i} \leqslant c^2 \sum_{i=2}^{k} \frac{1}{\{\log i\}\{\log(k+2-i)\}} = O\left(\frac{k}{\log^2 k}\right),$$

and, with (2.25),

$$kg_k = \sum_{j=1}^{k} g_j + O\left(\frac{k}{\log^2 k}\right).$$

This and (2.18) prove the theorem. ∎

THEOREM 8.

$$g_{k,N} = g_k + O\left(2^k \frac{\log^{2/3} N}{N}\right)$$

with an O-constant independent of k and N.

COROLLARY. $g_{k,N} = (1/\log k)\{1 + o(1)\}$ *if* $k \to \infty$ *and* $N \geqslant e^k$.

Proof of Theorem 8. The recursion formulas for $g_{k,N}$ in Theorem 6 and for g_k in (2.7) show

$$\begin{aligned} g_{k,N} - g_k &= k(\sigma_{k,N} - \sigma_k) - \sum_{i=1}^{k-1} (\sigma_{k-i,N} g_{i,N} - \sigma_{k-i} g_i) \\ &= k(\sigma_{k,N} - \sigma_k) \\ &\quad - \sum_{i=1}^{k-1} \{(\sigma_{k-i,N} - \sigma_{k-i}) g_i + \sigma_{k-i,N}(g_{i,N} - g_i)\}. \end{aligned}$$

By (2.12) and (2.14) we have $0 \leqslant g_k \leqslant 1$ for all $k \geqslant 0$, and this, in connection

with Lemma 2(2), gives

$$g_{k,N} - g_k = -\sum_{i=1}^{k-1} \sigma_{k-i,N}(g_{i,N} - g_i) + O\left(k\frac{\log^{2/3} N}{N}\right)$$

and further, because $0 \leqslant \sigma_{j,N} \leqslant 1$ [Lemma 2(1)]

$$|g_{k,N} - g_k| \leqslant \sum_{i=1}^{k-1} |g_{i,N} - g_i| + ck\frac{\log^{2/3} N}{N},$$

thereby proving the theorem inductively. ∎

The Corollary is an immediate consequence of Theorems 7 and 8.

THEOREM 9.

(1) $g_{k,N} \leqslant 1 - 1/N^k$ *for* $N \geqslant 1$, $k \geqslant 0$.
(2) $g_{k,N} \geqslant 0$ *if* $N \geqslant N_0$, $k \leqslant cN^{1/2}\log^{1/6} N \log_2^{-1/2} N$.
(3) $g_{k,N} \geqslant 0$ *if* $N \geqslant N_0$, $k \geqslant \{1.15/\tau(N)\} N \log^2 N$.

REMARK. The theorem states that for $N \geqslant N_0$ the left inequality of

$$1^k + 2^k + \cdots + (N-1)^k \leqslant \lambda_1^k + \lambda_2^k + \cdots + \lambda_{N-1}^k \leqslant 2^k + 3^k + \cdots + N^k$$

is always true, whereas the right one remains unproved for exponents k in an interval depending on N.

Figure 1 shows plots of $g_{k,N}$ against k for two values of N. I am indebted to H. Stuis for these graphs.

Proof. The recursion formula in Theorem 6 shows that for $k \geqslant 0$, similarly to (2.11),

$$g_{k+1,N} = \sum_{i=1}^{k} (\sigma_{k-i,N} - \sigma_{k+1-i,N}) g_{i,N} + D_{k+1,N}, \tag{2.26}$$

where

$$D_{k+1,N} = (k+1)\sigma_{k+1,N} - k\sigma_{k,N}. \tag{2.27}$$

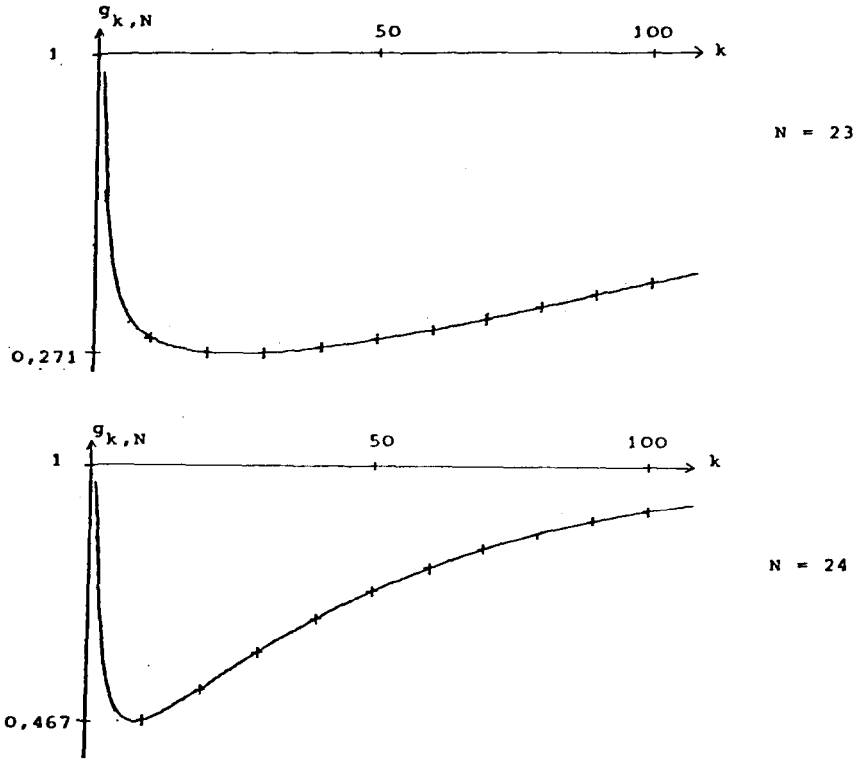


FIG. 1

$g_{0,N} = 0$ and $g_{1,N} = 1 - 1/N$ is clear. Now, if assertion (1) is valid for all exponents $\leqslant k$, then with (2.26) and because $(\sigma_{k,N})_{k \geqslant 0}$ is decreasing [Lemma 2(1)],

$$\begin{aligned} g_{k+1,N} &\leqslant (1 - \sigma_{k,N})\left(1 - \frac{1}{N^k}\right) + (k+1)\sigma_{k+1,N} - k\sigma_{k,N} \\ &= 1 - \frac{1}{N^k}(1 - \sigma_{k,N}) + (k+1)(\sigma_{k+1,N} - \sigma_{k,N}) \\ &\leqslant 1 - \frac{1}{N^k}(1 - \sigma_{k,N}) \\ &\leqslant 1 - \frac{1}{N^{k+1}}. \end{aligned}$$

(2): We cannot prove $g_{k,N} \geqslant 0$ from (2.26) in the same way as we have obtained $g_k > 0$ [i.e. (2.12)] from (2.11). The reason is that contrary to the differences $(k+1)\sigma_{k+1} - k\sigma_k$ in (2.13), the differences $D_{k+1,N}$ here are negative for certain indices.

Lemma 2(6) shows

$$D_{k+1,N} = \sigma_{k+1,N} - k(\sigma_{k,N} - \sigma_{k+1,N})$$

$$= \left\{1 - \left(1 - \frac{1}{N}\right)^k\right\}\sigma_{k+1,N}$$

$$+ O\left\{\frac{\log^{2/3} N}{N}\left(1 - \frac{1}{N}\right)^k\left(1 + \frac{k^2}{N^2}\right) + \frac{k}{2^k}(\log N)(\log_2 N)\right\},$$

and

$$\left(1 - \frac{1}{N}\right)^k\left(1 + \frac{k^2}{N^2}\right) = O(1),$$

$$\sigma_{k+1,N} = O\left(\frac{1}{k} + \frac{\log^{2/3} N}{N}\right),$$

$$0 \leqslant 1 - \left(1 - \frac{1}{N}\right)^k \leqslant \min\left\{1, \frac{k}{N}\right\}.$$

Hence, for all N and k,

$$D_{k+1,N} = O\left\{\frac{\log^{2/3} N}{N} + \frac{k}{2^k}(\log N)(\log_2 N)\right\}. \tag{2.28}$$

By (2.12) all g_k are positive; thus by Theorem 7

$$g_k \geqslant \frac{c_1}{\log(k+1)} \qquad \text{for} \quad k \geqslant 1 \text{ and } c_1 > 0.$$

Theorem 8 states

$$g_{k,N} = g_k + O\left(\frac{\log^{2/3} N}{N^{1-\log 2}}\right) \qquad \text{for} \quad k \leqslant \log N,$$

and hence

$$g_{k,N} \geqslant \frac{c_2}{\log(k+1)} \qquad \text{for} \quad N \geqslant N_1, \quad k \leqslant \log N, \tag{2.29}$$

if we choose $c_2 > 0$ small enough.

In the case $k > \log N$ we proceed inductively in two steps: If $\log N < k \leqslant N^{1/3}$ and if $g_{i,N} \geqslant 0$ for $i \leqslant k$, then $g_{1,N} = 1 - 1/N$, (2.26), (2.28), and Lemma 2 give

$$\begin{aligned} g_{k+1,N} &\geqslant \tfrac{1}{2}(\sigma_{k-1,N} - \sigma_{k,N}) + D_{k+1,N} \\ &\geqslant \frac{c_3}{k^2} + O\left\{ \frac{\log^{2/3} N}{N} + \frac{k}{2^k}(\log N)(\log_2 N) \right\}. \end{aligned}$$

This certainly is positive for $N \geqslant N_2$ as long as $k \leqslant N^{1/3}$. Finally, if $N^{1/3} < k \leqslant N/\log N$ and $N \geqslant N_3$, then—again under the assumption $g_{i,N} \geqslant 0$ for $i \leqslant k$—(2.26), (2.29), Lemma 2, and (2.28) show

$$\begin{aligned} g_{k+1,N} &\geqslant \sum_{1 \leqslant i \leqslant \log N} (\sigma_{k-i,N} - \sigma_{k+1-i,N}) g_{i,N} + D_{k+1,N} \\ &\geqslant c_4 \frac{\log N}{\log_2 N} \left\{ \frac{1}{k} \left(1 - \frac{1}{N}\right)^k \sigma_{k+1,N} + O\left(\frac{\log^{2/3} N}{kN} \right) \right\} + O\left(\frac{\log^{2/3} N}{N} \right) \\ &\geqslant c_5 \frac{\log N}{\log_2 N} \frac{1}{k^2} + O\left(\frac{\log^{2/3} N}{N} \right), \end{aligned}$$

which is positive as long as

$$k \leqslant c_6 N^{1/2} \log^{1/6} N \log_2^{-1/2} N.$$

To prove assertion (3) we use Theorem 11 of Section 3: In view of the fact that we have fairly well located in Theorem 5(3) the eigenvalues $\lambda_1, \ldots, \lambda_{N-1}$ of A_N with at most $2[\sqrt{N}]$ exceptions, equation (*) in Theorem 6 gives

$$\begin{aligned} g_{k,N} N^k &= \sum_{n=2}^{N} n^k - \sum_{n=1}^{N-1} \lambda_n^k \\ &\geqslant N^k - 2\sqrt{N} |\lambda_{\max}|^k. \end{aligned}$$

On the other hand Theorem 11(4) implies

$$1 \geqslant 2\sqrt{N}\left(\frac{\lambda_{\max}}{N}\right)^k$$

for $N \geqslant N_0$ and $k \geqslant \{1.15/\tau(N)\}N\log^2 N$, because $\frac{1}{2}\zeta_\infty < 1.15$. ■

3. THE LARGEST EIGENVALUE

THEOREM 10.

(1) $\sum_{k\geqslant 0}(\sigma_{k,N} - \sigma_{\infty,N}) = \zeta_\infty \log N + O(\log_2^2 N)$.
(2) *Let* $g^*_{k,N} = 1 - g_{k,N}$. *Then*

$$\sum_{k\geqslant 1} g^*_{k,N} = \frac{1}{\sigma_{\infty,N}} \sum_{k\geqslant 0}(\sigma_{k,N} - \sigma_{\infty,N})$$

$$= \frac{\zeta_\infty}{\tau(N)} N\log N\left\{1 + O\left(\frac{\log_2^2 N}{\log N}\right)\right\}$$

$$= O(N\log N).$$

Proof. (1):

$$\sum_{k\leqslant K}(\sigma_{k,N} - \sigma_{\infty,N}) = \sum_{k\leqslant K}\sigma_{k,N} + O\left(\frac{K}{N}\log_2 N\right) \quad \text{[Lemmata 2(4), 1(4)]}$$

$$= \sum_{k\leqslant K}\sigma_k + O\left(\frac{K}{N}\log^{2/3} N\right) \quad \text{[Lemma 2(2)]}$$

$$= \sum_{k\leqslant K}\frac{1}{k}\zeta_\infty + O\left(1 + \frac{K}{N}\log^{2/3} N\right) \quad \text{[Lemma 2(3)]}$$

$$= \zeta_\infty \log K + O\left(1 + \frac{K}{N}\log^{2/3} N\right),$$

$$\sum_{K<k\leqslant K'}(\sigma_{k,N}-\sigma_{\infty,N})=O\left((\log_2 N)\sum_{K<k\leqslant K'}\frac{1}{k}\right)\qquad\text{[Lemma 2(5)]}$$

$$=O\left((\log_2 N)\left(\log\frac{K'}{K}+\frac{1}{K}\right)\right),$$

$$\sum_{k>K'}(\sigma_{k,N}-\sigma_{\infty,N})=O\left(\sum_{k>K'}\left(1-\frac{1}{N}\right)^k\left(\frac{1}{k}+\frac{\log^{2/3}N}{N}\right)\right)$$

[Lemma 2(5)]

$$=O\left(\frac{N}{K'}\left(1-\frac{1}{N}\right)^{K'}+(\log^{2/3}N)\left(1-\frac{1}{N}\right)^{K'}\right).$$

Now we specify $K=N/\log N$, $K'=N\log N$ and add the equations.

(2): The transcription of (2.26) for $g^*_{k,N}$ instead of $g_{k,N}$ is

$$g^*_{k+1,N}=\sum_{i=1}^{k}(\sigma_{k-i,N}-\sigma_{k+1-i,N})g^*_{i,N}+(k+1)(\sigma_{k,N}-\sigma_{k+1,N}).\quad(3.1)$$

Then Theorems 6 and 9(1) show

$$0\leqslant g^*_{k,N}\leqslant N\left(\frac{\lambda_{\max}}{N}\right)^k,$$

and we have $|\lambda_{\max}/N|<1$ by Theorem 2. Hence the series in (2) is absolutely convergent, and (3.1) gives

$$\sum_{k\geqslant 0}g^*_{k+1,N}=\sum_{k\geqslant 0}\left(\sum_{i=1}^{k}(\sigma_{k-i,N}-\sigma_{k+1-i,N})g^*_{i,N}\right)$$

$$+\sum_{k\geqslant 0}(k+1)(\sigma_{k,N}-\sigma_{k+1,N})$$

$$=\sum_{i\geqslant 1}(1-\sigma_{\infty,N})g^*_{i,N}+\sum_{k\geqslant 1}k(\sigma_{k-1,N}-\sigma_{k,N});$$

thus

$$\sum_{k\geqslant 1}g^*_{k,N}=\frac{1}{\sigma_{\infty,N}}\sum_{k\geqslant 1}k(\sigma_{k-1,N}-\sigma_{k,N}).\quad(3.2)$$

Further we have

$$\sum_{k=1}^{K} k(\sigma_{k-1,N} - \sigma_{k,N}) = \sum_{k=0}^{K-1} (\sigma_{k,N} - \sigma_{\infty,N}) - K(\sigma_{K,N} - \sigma_{\infty,N})$$

$$= \sum_{k=0}^{K-1} (\sigma_{k,N} - \sigma_{\infty,N}) + O\left(K\left(1 - \frac{1}{N}\right)^K\right)$$

[Lemma 2(5)]

and hence

$$\sum_{k \geqslant 1} k(\sigma_{k-1,N} - \sigma_{k,N}) = \sum_{k \geqslant 0} (\sigma_{k,N} - \sigma_{\infty,N}).$$

This and (3.2) prove the first equation in (2) and, in connection with (1) and Lemma 2(4), also the second one. The third one then follows because $\tau(N) \geqslant 1$. ∎

THEOREM 11. *Let* $\lambda_{\max} = \max_{1 \leqslant n \leqslant N-1} |\lambda_n|$ *denote the spectral radius of* A_N.

(1) *There is exactly one zero of* $\chi_N(x)$ *in the interval* $[N-1, \infty[$, *and this is* $\lambda_{\max}$.

(2) $\lambda_{\max}$ *is a simple zero of* $\chi_N(x)$.

(3) $|\lambda| < \lambda_{\max}$ *for all zeros* $\lambda \neq \lambda_{\max}$ *of* $\chi_N(x)$.

(4) $$\lambda_{\max} = N - \frac{\tau(N)}{\zeta_\infty \log N}\left\{1 + O\left(\frac{\log_2^2 N}{\log N}\right)\right\}.$$

COROLLARY.

(1) $$\sum_{k \geqslant 1} \frac{1}{k} g^*_{k,N} = \log \frac{1}{\sigma_{\infty,N}}.$$

(2) $$\chi_N(N) = \frac{(N-1)!}{N} \tau(N).$$

Proof. The radii of convergence of the series

$$S_N(x) = \sum_{k \geqslant 0} \sigma_{k,N} x^k,$$

$$G_N(x) = \sum_{k \geqslant 0} g_{k+1,N} x^k, \qquad G^*_N(x) = \sum_{k \geqslant 0} g^*_{k+1,N} x^k,$$

$$D_N(x) = \sum_{k \geqslant 1} (\sigma_{k-1,N} - \sigma_{k,N}) x^k = 1 - (1-x) S_N(x)$$

are

$$\rho_{S_N(x)} = 1, \tag{3.3}$$

$$\rho_{G_N(x)} = 1, \tag{3.4}$$

$$\rho_{G_N^*(x)} = \frac{N}{\lambda_{\max}}, \tag{3.5}$$

$$\rho_{D_N(x)} = 1 + \frac{1}{N-1}. \tag{3.6}$$

(3.3) is valid because according to Lemma 2(1), $(\sigma_{k,N})_{k\geqslant 0}$ converges to $\sigma_{\infty,N} > 0$. The remark following Theorem 6 says $\lim_{k\to\infty} g_{k,N} = 1$, which proves (3.4). From Equation (*) in Theorem 6 we obtain

$$g_{k,N}^* = \sum_{n=1}^{N-1} \left(\frac{\lambda_n}{N}\right)^k + O\left(N\left(1-\frac{1}{N}\right)^k\right),$$

and in view of $\lambda_{\max} > N-1$ the Hadamard formula for the radius of convergence gives (3.5).

Lemma 2(5) shows

$$0 \leqslant \sigma_{k-1,N} - \sigma_{k,N} \leqslant c\left(1-\frac{1}{N}\right)^k,$$

and hence in any case

$$\rho_{D_N(x)} \geqslant 1 + \frac{1}{N-1}. \tag{3.7}$$

Further we have the identities

$$G_N(x)S_N(x) = S_N'(x), \tag{3.8}$$

$$G_N^*(x) = \frac{1}{1-x} - G_N(x), \tag{3.9}$$

$$G_N^*(x) = \frac{D_N'(x)}{1-D_N(x)}. \tag{3.10}$$

(3.8) is a consequence of the recursion formula in Theorem 6, (3.9) comes from $g^*_{k,N} = 1 - g_{k,N}$, and both together give

$$G^*_N(x) = \frac{1}{1-x} - \frac{S'_N(x)}{S_N(x)} = \frac{S_N(x) - (1-x)S'_N(x)}{(1-x)S_N(x)} = \frac{D'_N(x)}{1 - D_N(x)}.$$

If $x \geqslant 0$ increases, so does $D_N(x)$; in particular

$$D_N(1) = 1 - \sigma_{\infty,N}. \tag{3.11}$$

Let x^* denote that positive argument such that $D_N(x^*) = 1$. Then (3.10) shows

$$x^* = \rho_{G^*_N(x)};$$

hence in view of (3.5)

$$D_N\left(\frac{N}{\lambda_{\max}}\right) = 1. \tag{3.12}$$

We integrate (3.10) from 0 to y, $y < N/\lambda_{\max}$, and obtain

$$\sum_{k \geqslant 1} \frac{1}{k} g^*_{k,N} y^k = -\log\{1 - D_N(y)\}. \tag{3.13}$$

The substitution $y = 1$ and (3.11) prove the first assertion of the Corollary.

Theorem 6 says

$$\sum_{n=1}^{N-1} \left(\frac{\lambda_n}{y}\right)^k = g^*_{k,N}\left(\frac{N}{y}\right)^k + \sum_{n=2}^{N-1}\left(\frac{n}{y}\right)^k.$$

We take $|y| > \lambda_{\max}$, add these equations for all $k \geqslant 0$, and divide by y:

$$\sum_{n=1}^{N-1} \frac{1}{y - \lambda_n} = \sum_{k \geqslant 0} g^*_{k,N} \frac{N^k}{y^{k+1}} + \sum_{n=2}^{N-1} \frac{1}{y - n}. \tag{3.14}$$

By integration we get

$$\sum_{n=1}^{N-1} \log(y - \lambda_n) = \log y - \sum_{k \geqslant 1} \frac{g^*_{k,N}}{k}\left(\frac{N}{y}\right)^k + \sum_{n=2}^{N-1} \log(y - n) + c_1,$$

and finally

$$\prod_{n=1}^{N-1}(y-\lambda_n)=c_2 y\prod_{n=2}^{N-1}(y-n)\exp\left\{-\sum_{k\geqslant 1}\frac{g^*_{k,N}}{k}\left(\frac{N}{y}\right)^k\right\}.$$

$y\to\infty$ says $c_2=1$, and with (3.13) we conclude

$$\chi_N(x)=x\left\{1-D_N\left(\frac{N}{x}\right)\right\}\prod_{n=2}^{N-1}(x-n), \tag{3.15}$$

provided $x>N-1$, which comes from (3.7). But Theorem 4 says $\lim_{x\downarrow N-1}\chi_N(x)\neq 0$, hence (3.15) shows

$$\lim_{x\downarrow N-1} D_N\left(\frac{N}{x}\right)=\infty,$$

which in connection with (3.7) proves (3.6).

We substitute $x=N$ in (3.15) and conclude with (3.11) and Lemma 2(4)

$$\chi_N(N)=N(N-2)!\{1-D_N(1)\}=N(N-2)!\sigma_{\infty,N}=\frac{(N-1)!}{N}\tau(N),$$

which is the second assertion of the Corollary.

(1): $D_N(x)$ increases for $x>0$; hence (3.15) and (3.12) show that $\chi_N(x)$ has just one zero in $]N-1,\infty[$, which is $\lambda_{\max}$.

(2): Differentiation of (3.15) gives

$$\chi'_N(\lambda_{\max})=D'_N\left(\frac{N}{\lambda_{\max}}\right)\frac{N}{\lambda_{\max}}\prod_{n=2}^{N-1}(\lambda_{\max}-n)\neq 0,$$

for $D'_N(x)$ is positive if $x\geqslant 0$. Hence $\lambda_{\max}$ is a zero of $\chi_N(x)$ of order 1.

(3): (1) says that $\lambda_{\max}$ is the largest positive zero of $\chi_N(x)$. Further we have $\lambda_{\max}>N-1$, which implies that $-\lambda_{\max}$ cannot be a zero of $\chi_N(x)$ by Theorem 2. Assume finally that $\lambda=a+bi$, $b\neq 0$, is a zero of modulus $\lambda_{\max}$. Then $D_N(N/\lambda)=1$ by (3.15), which is impossible, because we have $D_N(N/\lambda_{\max})=1$ and all coefficients of the series $D_N(x)$ are nonnegative.

(4): $D'_N(x)$ increases; hence the mean-value theorem shows for $1+\varepsilon = N/\lambda_{\max}$

$$\frac{1}{\varepsilon}\{D_N(1+\varepsilon) - D_N(1)\} \geqslant D'_N(1).$$

This implies

$$\begin{aligned}
\varepsilon &\leqslant \frac{1}{D'_N(1)}\left\{D_N\left(\frac{N}{\lambda_{\max}}\right) - D_N(1)\right\} \\
&\leqslant \frac{\sigma_{\infty,N}}{\zeta_\infty \log N + O\left(\log_2^2 N\right)} \qquad [(3.11), (3.12), \text{Theorem } 10(1)] \\
&\leqslant \frac{\tau(N)}{\zeta_\infty N \log N}\left\{1 + O\left(\frac{\log_2^2 N}{\log N}\right)\right\} \qquad [\text{Lemma } 2(4)],
\end{aligned}$$

and thus

$$\lambda_{\max} \geqslant N - \frac{\tau(N)}{\zeta_\infty \log N}\left\{1 + O\left(\frac{\log_2^2 N}{\log N}\right)\right\}. \tag{3.16}$$

On the other hand (3.14) with $y = N$ is

$$\frac{1}{N}\sum_{k\geqslant 0} g^*_{k,N} = \sum_{n=1}^{N-1} \frac{1}{N-\lambda_n} - \sum_{n=2}^{N-1} \frac{1}{N-n},$$

and with Theorem 10(2) we obtain

$$\frac{\zeta_\infty}{\tau(N)} \log N\left\{1 + O\left(\frac{\log_2^2 N}{\log N}\right)\right\} = \sum_{n=1}^{N-1} \frac{1}{N-\lambda_n} - \sum_{m=1}^{N-2} \frac{1}{m}.$$

The pairs $a \pm bi$ of complex eigenvalues fulfill

$$\frac{1}{N-\lambda} + \frac{1}{N-\bar{\lambda}} = 2\frac{N-a}{(N-a)^2+b^2} > 0,$$

for $|\lambda| < N$ implies $|a| < N$. Hence

$$\frac{\zeta_\infty}{\tau(N)} \log N \left\{ 1 + O\left(\frac{\log_2^2 N}{\log N} \right) \right\} \geqslant \sum_{\substack{1 \leqslant n \leqslant N-1 \\ \lambda_n \text{ real}}} \frac{1}{N - \lambda_n} - \sum_{m=1}^{N-2} \frac{1}{m}.$$

According to Theorem 5(2), each interval $[n, n+1[$, $N/2 < n \leqslant N-2$, contains an eigenvalue of A_N, and it follows that

$$\sum_{\substack{1 \leqslant n \leqslant N-1 \\ \lambda_n \text{ real}}} \frac{1}{N - \lambda_n} \geqslant \frac{1}{N - \lambda_{\max}} + \sum_{2 \leqslant m < N/2} \frac{1}{m} + O\left(\frac{1}{N} \right).$$

Now we can conclude

$$\frac{\zeta_\infty}{\tau(N)} \log N \left\{ 1 + O\left(\frac{\log_2^2 N}{\log N} \right) \right\} \geqslant \frac{1}{N - \lambda_{\max}} - c,$$

and in view of Lemma 1(4) the constant c disappears within the O-term on the left, so that we finally get

$$N - \lambda_{\max} \geqslant \frac{\tau(N)}{\zeta_\infty \log N} \left\{ 1 + O\left(\frac{\log_2^2 N}{\log N} \right) \right\},$$

which in connection with (3.16) proves the last assertion of the theorem. ∎

APPENDIX

Proof of Lemma 1.

(1) is an immediate consequence of the definition of $\tau_{k,N}$.

(2): $\tau_{k+1,N} \geqslant \tau_{k,N}$ reads

$$\frac{1}{N^k} \sum_{\substack{1 \leqslant i_1 \leqslant \cdots \leqslant i_k \leqslant N \\ i_1 \mid \cdots \mid i_k \mid N}} i_1 \cdots i_k \geqslant \frac{1}{N^{k-1}} \sum_{\substack{1 \leqslant i_1 \leqslant \cdots \leqslant i_{k-1} \leqslant N \\ i_1 \mid \cdots \mid i_{k-1} \mid N}} i_1 \cdots i_{k-1},$$

and this is valid because every chain $i_1 \mid \cdots \mid i_{k-1}$ on the right is part of a chain $i_1 \mid \cdots \mid i_{k-1} \mid i_k = N$ on the left.

(3):

$$\tau_{k,p^s} = \frac{1}{p^{s(k-1)}}\left\{ \sum_{\substack{1<i_1\leqslant\cdots\leqslant i_{k-1}\leqslant p^s \\ i_1\mid\cdots\mid i_{k-1}\mid p^s}} i_1\cdots i_{k-1} + \sum_{\substack{1=i_1\leqslant i_2\leqslant\cdots\leqslant i_{k-1}\leqslant p^s \\ i_2\mid\cdots\mid i_{k-1}\mid p^s}} i_2\cdots i_{k-1}\right\}$$

$$= \frac{1}{p^{s(k-1)}}\left\{ p^{k-1} \sum_{\substack{1\leqslant j_1\leqslant\cdots\leqslant j_{k-1}\leqslant p^{s-1} \\ j_1\mid\cdots\mid j_{k-1}\mid p^{s-1}}} j_1\cdots j_{k-1} + p^{s(k-2)}\tau_{k-1,p^s}\right\}$$

$$= \tau_{k,p^{s-1}} + \frac{1}{p^s}\tau_{k-1,p^s}.$$

The second equation then follows by induction on s.

(4): The first equation follows directly from (3) if $N = p^s$ is a prime power, and then we use (1). The second equation is a consequence of $\sum_{p\leqslant M} 1/p = O(\log_2 M)$.

(5): Equations (3) and (4) show for prime powers

$$\tau(p^s) - \tau_{k,p^s} = O\left(\frac{1}{p^k}\right).$$

Further, $\tau(N)$ and $\tau_{k,N}$ are multiplicative in N and always $\geqslant 1$; hence in view of (2) and (4) we obtain for $k \geqslant 2$

$$0 \leqslant \tau(N) - \tau_{k,N} \leqslant \tau(N) \sum_{p^s \,\|\, N} \left\{\tau(p^s) - \tau_{k,p^s}\right\}$$

$$= O\left(\log_2 N \sum_{p^s \,\|\, N} \frac{1}{p^k}\right) = O\left(\frac{\log_2 N}{2^k}\right). \qquad \blacksquare$$

Proof of Lemma 2. (1): Clearly $\sigma_{1,N} = 1 - 1/N < 1 = \sigma_{0,N}$, and

$$N^{k+1}\sigma_{k+1,N} = \sum_{\substack{2 \leqslant i_1 \leqslant \cdots \leqslant i_{k+1} \leqslant N \\ i_1 | \cdots | i_{k+1}}} i_1 \cdots i_k$$

$$= \sum_{\substack{2 \leqslant i_1 \leqslant \cdots \leqslant i_k \leqslant N \\ i_1 | \cdots | i_k}} i_1 \cdots i_k \left[\frac{N}{i_k}\right]$$

$$\leqslant N \sum_{\substack{2 \leqslant i_1 \leqslant \cdots \leqslant i_k \leqslant N \\ i_1 | \cdots | i_k}} i_1 \cdots i_{k-1} = N^{k+1}\sigma_{k,N}.$$

(2): We have $\sigma_{1,N} = 1 - 1/N$ and $\sigma_1 = 1$, so we can assume $k \geqslant 2$. First we define, for real $x \geqslant 1$ and natural numbers k,

$$\sigma^*_{k,x} := \frac{1}{x^k} \sum_{\substack{1 \leqslant i_1 \leqslant \cdots \leqslant i_k \leqslant x \\ i_1 | \cdots | i_k}} i_1 \cdots i_{k-1}.$$

It is sufficient to prove

$$\sigma^*_{k,x} = \sigma_k + O\left(\frac{\log^{2/3}(2x)}{x}\right). \tag{4.1}$$

For this implies

$$\sigma_{k,N} = \sigma^*_{k,N} - \frac{1}{N}\sigma^*_{k-1,N} = \sigma_k + O\left(\frac{\log^{2/3} N}{N}\right).$$

If $k = 1$, then $\sigma^*_{1,x} = [x]/x = 1 + O(1/x)$. If $k = 2$, (4.1) says

$$\frac{1}{x^2} \sum_{1 \leqslant i_1} i_1 \left[\frac{x}{i_1}\right] = \tfrac{1}{2}\zeta(2) + O\left(\frac{\log^{2/3}(2x)}{x}\right), \tag{4.2}$$

which is valid by [5, p. 99, Satz 4]. If $k \geqslant 3$, we use the estimate

$$\sum_{m,n \geqslant 1} m^k n^{k-1} \left[\frac{x}{mn}\right] = \frac{1}{k+1}\zeta(2)\zeta(k+1)x^{k+1} + O\left(x^k \log^{2/3}(x+1)\right), \tag{4.3}$$

which will be shown for $k \geqslant 2$ and $x > 0$ and which implies:

$$x^k \sigma^*_{k,x} = \sum_{\substack{1 \leqslant i_1 \leqslant \cdots \leqslant i_{k-1} \leqslant x \\ i_1 | \cdots | i_{k-1}}} i_1 \cdots i_{k-1} \left[\frac{x}{i_{k-1}} \right]$$

$$= \sum_{n_1, \ldots, n_{k-1} \geqslant 1} n_1^{k-1} \cdots n_{k-1}^1 \left[\frac{x}{n_1 \cdots n_{k-1}} \right]$$

$$= \sum_{n_3, \ldots, n_{k-1} \geqslant 1} n_3^{k-3} \cdots n_{k-1}^1 \left\{ \sum_{n_1, n_2 \geqslant 1} n_1^{k-1} n_2^{k-2} \left[\frac{x/(n_3 \cdots n_{k-1})}{n_1 n_2} \right] \right\}$$

$$= \sum_{n_3, \ldots, n_{k-1} \geqslant 1} n_3^{k-3} \cdots n_{k-1}^1 \left\{ \frac{1}{k} \zeta(2) \zeta(k) \left(\frac{x}{n_3 \cdots n_{k-1}} \right)^k \right.$$

$$\left. + O\left(\left(\frac{x}{n_3 \cdots n_{k-1}} \right)^{k-1} \log^{2/3}(2x) \right) \right\}$$

$$= \frac{1}{k} \zeta(2) \cdots \zeta(k) x^k + O\left(\zeta_\infty x^{k-1} \log^{2/3}(2x) \right).$$

It remains to prove (4.3):

$$\sum_{m,n \geqslant 1} m^k n^{k-1} \left[\frac{x}{mn} \right] = \sum_{1 \leqslant n \leqslant x} n^{k-1} \left(\sum_{1 \leqslant m \leqslant x/n} m^k \left[\frac{x}{mn} \right] \right)$$

$$= \sum_{1 \leqslant n \leqslant x} n^{k-1} \left\{ \sum_{1 \leqslant r \leqslant x/n} \left(\sum_{1 \leqslant m \leqslant x/nr} m^k \right) \right\}.$$

By [2, p. 187] the inner sum is

$$\sum_{1 \leqslant m \leqslant x/nr} m^k = \frac{1}{k+1} \left(\frac{x}{nr} \right)^{k+1} - \psi\left(\frac{x}{nr} \right) \left(\frac{x}{nr} \right)^k$$

$$+ k \int_1^{x/nr} \psi(u) u^{k-1} \, du + O(1),$$

where $\psi(y) = y - [y] - \frac{1}{2}$. We substitute and obtain the following four sums:

$$S_1 = \sum_{1 \leqslant n \leqslant x} n^{k-1} \sum_{1 \leqslant r \leqslant x/n} \frac{1}{k+1} \left(\frac{x}{nr}\right)^{k+1}$$

$$= \frac{x^{k+1}}{k+1} \sum_{1 \leqslant n \leqslant x} \frac{1}{n^2} \left(\sum_{1 \leqslant r \leqslant x/n} \frac{1}{r^{k+1}} \right)$$

$$= \frac{x^{k+1}}{k+1} \sum_{1 \leqslant n \leqslant x} \frac{1}{n^2} \left\{ \zeta(k+1) + O\left(\left(\frac{n}{x}\right)^k \right) \right\}$$

$$= \frac{1}{k+1} \zeta(2)\zeta(k+1) x^{k+1} + O(x^k),$$

$$S_2 = \sum_{1 \leqslant n \leqslant x} n^{k-1} \sum_{1 \leqslant r \leqslant x/n} \psi\left(\frac{x}{nr}\right) \left(\frac{x}{nr}\right)^k$$

$$= x^k \sum_{1 \leqslant r \leqslant x} \frac{1}{r^k} \sum_{1 \leqslant n \leqslant x/r} \frac{1}{n} \psi\left(\frac{x/r}{n}\right)$$

$$= x^k \sum_{1 \leqslant r \leqslant x} \frac{1}{r^k} O(\log^{2/3}(x+1)) \qquad [5, \text{p. } 88]$$

$$= O(x^k \log^{2/3}(x+1)),$$

$$S_3 = \sum_{1 \leqslant n \leqslant x} n^{k-1} \sum_{1 \leqslant r \leqslant x/n} k \int_1^{x/nr} \psi(u) u^{k-1} \, du$$

$$= k \sum_{1 \leqslant n \leqslant x} n^{k-1} \int_0^{x/n} \psi(u) u^{k-1} \, du$$

$$+ k \sum_{1 \leqslant n \leqslant x} n^{k-1} \sum_{2 \leqslant r \leqslant x/n} \int_1^{x/nr} \psi(u) u^{k-1} \, du + O(x^k)$$

$$= k \sum_{1 \leqslant n \leqslant x} n^{k-1} \int_0^x \frac{1}{n} \psi\left(\frac{v}{n}\right) \left(\frac{v}{n}\right)^{k-1} dv$$

$$+ k \sum_{1 \leqslant n \leqslant x} n^{k-1} \sum_{2 \leqslant r \leqslant x/n} O\left(\left(\frac{x}{nr}\right)^{k-1} \right) + O(x^k)$$

$$= k \int_1^x \left\{ \sum_{1 \leqslant n \leqslant v} \frac{1}{n} \psi\left(\frac{v}{n}\right) + \sum_{v < n \leqslant x} \frac{1}{n} \psi\left(\frac{v}{n}\right) \right\} v^{k-1} \, dv + O(x^k),$$

and, because the first sum is $O(\log^{2/3}(x+1))$ (again by [5, p. 88]) and the second one is $O(\Sigma_{v<n\leqslant x} 1/n)$, S_3 is

$$\begin{aligned} &= k\left(\int_1^x v^{k-1}\,dv\right)O\left(\log^{2/3}(x+1)\right) \\ &\quad + k\int_1^x \left\{\log x - \log v + O\left(\frac{1}{v}\right)\right\} v^{k-1}\,dv + O(x^k) \\ &= O\left(x^k \log^{2/3}(x+1)\right). \end{aligned}$$

$$S_4 = \sum_{1\leqslant n\leqslant x} n^{k-1} \sum_{1\leqslant r\leqslant x/n} O(1) = O(x^k).$$

Taking all sums together, we obtain (4.3).

(3):

$$\begin{aligned} \zeta_\infty - \prod_{s=2}^{k} \zeta(s) &= \prod_{s=2}^{k} \zeta(s)\left(\prod_{s>k} \zeta(s) - 1\right) \\ &= O\left(\prod_{s>k}\left(1+\frac{c}{2^s}\right) - 1\right) \\ &= O\left(\exp\left(c\sum_{s>k} 2^{-s}\right) - 1\right) = O\left(\frac{1}{2^k}\right). \end{aligned}$$

(4) follows from (1) and Lemma 1(5), when we observe that

$$\begin{aligned} \sigma_{k,N} &= \frac{1}{N^k}\left\{ \sum_{\substack{2\leqslant i_1\leqslant\cdots\leqslant i_k\leqslant N-1 \\ i_1\mid\cdots\mid i_k}} i_1\cdots i_{k-1} + \sum_{\substack{2\leqslant i_1\leqslant\cdots\leqslant i_k = N \\ i_1\mid\cdots\mid i_k}} i_1\cdots i_{k-1}\right\} \\ &= \left(1-\frac{1}{N}\right)^k \sigma_{k,N-1} + \frac{1}{N}\tau_{k,N} - \frac{1}{N^2}\tau_{k-1,N}. \end{aligned} \tag{4.4}$$

(5): In connection with (1), (2), (4), and Lemma 1, Equation (4.4) implies

$$\begin{aligned} 0 \leqslant \sigma_{k,N} - \sigma_{\infty,N} &\leqslant \left(1-\frac{1}{N}\right)^k \sigma_{k,N-1} - \frac{1}{N}\left(\tau(N) - \tau_{k,N}\right) \\ &\quad + \frac{1}{N^2}\left(\tau(N) - \tau_{k-1,N}\right) \\ &\leqslant \left(1-\frac{1}{N}\right)^k \left\{\frac{\zeta_\infty}{k} + O\left(\frac{\log^{2/3} N}{N}\right)\right\} + \frac{1}{N^2} O\left(\frac{\log_2 N}{2^k}\right), \end{aligned} \tag{4.5}$$

which proves the first estimate in (5). Equation (4.5) in connection with (4) and Lemma 1 shows

$$0 \leqslant \sigma_{k,N} - \sigma_{\infty,N} \leqslant \left(1 - \frac{1}{N}\right)^k (\sigma_{k,N-1} - \sigma_{\infty,N-1}) + c\left(1 - \frac{1}{N}\right)^k \frac{\log_2(3N)}{N} + c\frac{\log_2(3N)}{N^2 2^k}$$

and hence successively

$$\sigma_{k,N} - \sigma_{\infty,N} \leqslant c\log_2(3N)\left\{\sum_{n=1}^{N-1} n^{k-1}\right\}\frac{1}{N^k} + \frac{c}{2^k} \leqslant c\frac{\log_2(3N)}{k}.$$

(6): From (4.4) we obtain with Lemma 1(5)

$$\sigma_{k,N} - \sigma_{k+1,N} = \left(1 - \frac{1}{N}\right)^k (\sigma_{k,N-1} - \sigma_{k+1,N-1}) + \left(1 - \frac{1}{N}\right)^k \frac{1}{N}\sigma_{k+1,N\ 1} + O\left(\frac{\log_2 N}{2^k N}\right).$$

Induction on N then gives

$$\sigma_{k,N} - \sigma_{k+1,N} = \frac{1}{N^k}\sum_{m=2}^{N-1}\frac{m^k}{m+1}\sigma_{k+1,m} + O\left(\frac{1}{2^k}(\log N)(\log_2 N)\right)$$

$$= \sigma_{k+1,N}\frac{1}{N^k}\sum_{m=2}^{N-1}\frac{m^k}{m+1} + \frac{1}{N^k}\sum_{m=2}^{N-1}\frac{m^k}{m+1}(\sigma_{k+1,m} - \sigma_{k+1,N}) + O\left(\frac{1}{2^k}(\log N)(\log_2 N)\right).$$

Using

$$\sum_{m=1}^{N-1} m^{k-1} = \frac{1}{k}(N-1)^k + O\big((N+k)(N-1)^{k-2}\big) \qquad [2, \text{p. } 187],$$

$$\sigma_{k+1,m} - \sigma_{k+1,N} = O\left(\frac{\log^{2/3} m}{m}\right) \qquad [\text{by (2)}],$$

and

$$\sigma_{k+1,N} = O\left(\frac{1}{k} + \frac{\log^{2/3} N}{N}\right) \qquad [\text{by (2) and (3)}],$$

we finally conclude

$$\sigma_{k,N} - \sigma_{k+1,N} = \frac{1}{k}\left(1 - \frac{1}{N}\right)^k \sigma_{k+1,N} + O\left\{\frac{\log^{2/3} N}{kN}\left(1 - \frac{1}{N}\right)^k \left(1 + \frac{k^2}{N^2}\right) + \frac{1}{2^k}(\log N)(\log_2 N)\right\}. \qquad \blacksquare$$

REFERENCES

1 M. Davis, Y. Matijasevič, and J. Robinson, Hilbert's tenth problem. Diophantine equations: Positive aspects of a negative solution, *Proc. Sympos. Pure Math.* XXVIII:323–378 (1976).

2 F. Fricker, *Einführung in die Gitterpunktlehre*, Birkhäuser, Basel, 1982.

3 W. Schwarz, *Einführung in Methoden und Ergebnisse der Primzahltheorie*, Bibliographisches Institut, Mannheim, 1969.

4 E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, Clarendon, Oxford, 1951.

5 A. Walfisz, *Weylsche Exponentialsummen in der neueren Zahlentheorie*, Deutscher Verlag der Wissenschaften, Berlin, 1963.

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